```
#include <iostream>
#include <string>
using namespace std;

// Node class for Music Playlist
class Node {
public:
    string songTitle;
    string artistName;

    Node* prev;
    Node* next;

    Node(string title, string artist) {
        songTitle = title;
        artistName = artist;
        prev = nullptr;
        next = nullptr;
    }
};

// Doubly Linked List for Music Playlist
class MusicPlaylist {
private:
    Node* head;       // head points to the first song
    Node* current;    // current points to the currently selected/playing song

public:

    // Constructor
    MusicPlaylist() {
        head = nullptr;
        current = nullptr;
    }

    // Add a new song at the end of the playlist
    void addSong(string title, string artist) {

        // p points to the newly created node
        Node* p = new Node(title, artist);

        // If playlist is empty
```

```
    if (head == nullptr) {
        head = p;
        current = head;

        cout << "Song added successfully.\n";
        return;
    }

    // q is used as the traversal pointer
    Node* q = head;

    // Move q to the last node
    while (q->next != nullptr) {
        q = q->next;
    }

    // Connect the new node
    q->next = p;
    p->prev = q;

    cout << "Song added successfully.\n";
}

// Delete a song using song title
void deleteSong(string title) {

    if (head == nullptr) {
        cout << "Playlist is empty. Nothing to delete.\n";
        return;
    }

    // q is used to search for the song
    Node* q = head;

    // Search for the song
    while (q != nullptr && q->songTitle != title) {
        q = q->next;
    }

    // Song not found
    if (q == nullptr) {
        cout << "Song not found in the playlist.\n";
```

```
    return;
}

// p points to the node that will be deleted
Node* p = q;

// Case 1: deleting the only song
if (p == head && p->next == nullptr) {
    head = nullptr;
    current = nullptr;

    delete p;

    cout << "Song deleted successfully.\n";
    return;
}

// Case 2: deleting the first song
if (p == head) {
    head = p->next;
    head->prev = nullptr;

    // If current was the deleted song
    if (current == p) {
        current = head;
    }

    delete p;

    cout << "Song deleted successfully.\n";
    return;
}

// Case 3: deleting a middle or last song

if (p->prev != nullptr) {
    p->prev->next = p->next;
}

if (p->next != nullptr) {
    p->next->prev = p->prev;
}
```

```
    // If current was the deleted song
    if (current == p) {

        // Move current to next song if possible
        if (p->next != nullptr)
            current = p->next;
        else
            current = p->prev;
    }

    delete p;

    cout << "Song deleted successfully.\n";
}

// Display playlist from first song to last song
void displayPlaylist() {

    if (head == nullptr) {
        cout << "Playlist is empty.\n";
        return;
    }

    // q is used for forward traversal
    Node* q = head;

    cout << "\n--- Playlist (Forward) ---\n";

    while (q != nullptr) {
        cout << "Song: " << q->songTitle
             << " | Artist: " << q->artistName;

        if (q == current)
            cout << "  <-- CURRENT";

        cout << endl;

        q = q->next;
    }
}
```

```
// Display playlist from last song to first song
void displayReverse() {

    if (head == nullptr) {
        cout << "Playlist is empty.\n";
        return;
    }

    // q is initially at head
    Node* q = head;

    // Move q to the last node
    while (q->next != nullptr) {
        q = q->next;
    }

    cout << "\n--- Playlist (Reverse) ---\n";

    // Traverse using prev pointer
    while (q != nullptr) {

        cout << "Song: " << q->songTitle
             << " | Artist: " << q->artistName;

        if (q == current)
            cout << "  <-- CURRENT";

        cout << endl;

        q = q->prev;
    }
}

// Play the current song
void playCurrentSong() {

    if (current == nullptr) {
        cout << "No song is currently selected.\n";
        return;
    }

    cout << "\nNow Playing: "
```

```
        << current->songTitle
        << " - "
        << current->artistName
        << endl;
}

// Move current pointer to the next song
void playNextSong() {

    if (current == nullptr) {
        cout << "Playlist is empty.\n";
        return;
    }

    // current moves using the next pointer
    if (current->next != nullptr) {

        current = current->next;

        cout << "Moved to next song.\n";
        playCurrentSong();
    }
    else {
        cout << "Already at the last song. No next song available.\n";
    }
}

// Move current pointer to the previous song
void playPreviousSong() {

    if (current == nullptr) {
        cout << "Playlist is empty.\n";
        return;
    }

    // current moves using the prev pointer
    if (current->prev != nullptr) {

        current = current->prev;

        cout << "Moved to previous song.\n";
        playCurrentSong();
```

```
    }
    else {
        cout << "Already at the first song. No previous song available.\n";
    }
}

// Play the complete playlist once
void playCompletePlaylist() {

    if (head == nullptr) {
        cout << "Playlist is empty.\n";
        return;
    }

    cout << "\n--- Playing Complete Playlist ---\n";

    // q starts from head
    Node* q = head;

    while (q != nullptr) {

        cout << "Playing: "
             << q->songTitle
             << " - "
             << q->artistName
             << endl;

        q = q->next;
    }

    // After complete playback, current remains at its selected song.
}

// Repeat the complete playlist specified number of times
void repeatAll() {

    if (head == nullptr) {
        cout << "Playlist is empty.\n";
        return;
    }

    int repetitions;
```

```
    cout << "Enter number of repetitions: ";
    cin >> repetitions;

    if (repetitions <= 0) {
        cout << "Invalid number of repetitions.\n";
        return;
    }

    cout << "\n--- Repeat All ---\n";

    for (int i = 1; i <= repetitions; i++) {

        cout << "\nRepetition " << i << ":\n";

        // Restart playback from head
        Node* q = head;

        while (q != nullptr) {

            cout << "Playing: "
                 << q->songTitle
                 << " - "
                 << q->artistName
                 << endl;

            q = q->next;
        }
    }
}

// Show currently selected song
void showCurrentSong() {

    if (current == nullptr) {
        cout << "No current song selected.\n";
        return;
    }

    cout << "\nCurrent Song:\n";
    cout << "Title  : " << current->songTitle << endl;
    cout << "Artist : " << current->artistName << endl;
```

```
    }
};


// Main function
int main() {

    MusicPlaylist playlist;

    int choice;
    string title;
    string artist;

    do {

        cout << "\n====================================\n";
        cout << "       MUSIC PLAYLIST MANAGER\n";
        cout << "====================================\n";

        cout << "1. Add Song\n";
        cout << "2. Delete Song\n";
        cout << "3. Display Playlist\n";
        cout << "4. Display Playlist in Reverse\n";
        cout << "5. Play Current Song\n";
        cout << "6. Play Next Song\n";
        cout << "7. Play Previous Song\n";
        cout << "8. Play Complete Playlist\n";
        cout << "9. Repeat All\n";
        cout << "10. Show Current Song\n";
        cout << "11. Exit\n";

        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {

        case 1:

            cin.ignore();

            cout << "Enter song title: ";
            getline(cin, title);
```

```
            cout << "Enter artist name: ";
            getline(cin, artist);

            playlist.addSong(title, artist);

            break;


        case 2:

            cin.ignore();

            cout << "Enter song title to delete: ";
            getline(cin, title);

            playlist.deleteSong(title);

            break;


        case 3:

            playlist.displayPlaylist();

            break;


        case 4:

            playlist.displayReverse();

            break;


        case 5:

            playlist.playCurrentSong();

            break;
```

```
case 6:
    playlist.playNextSong();
    break;

case 7:
    playlist.playPreviousSong();
    break;

case 8:
    playlist.playCompletePlaylist();
    break;

case 9:
    playlist.repeatAll();
    break;

case 10:
    playlist.showCurrentSong();
    break;

case 11:
    cout << "Exiting Music Playlist Manager...\n";
    break;
```

```
        default:

            cout << "Invalid choice. Please try again.\n";
        }

    } while (choice != 11);

    return 0;
}



/*
OUTPUT
PS C:\Users\Admin\OneDrive\Desktop\DSAcce2> ./musicplaylist

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 1
Enter song title: Kesariya
Enter artist name: Arjit
Song added successfully!

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
```

9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 1
Enter song title: Heeriye
Enter artist name: Jasleen
Song added successfully!

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 1
Enter song title: Zingat
Enter artist name: Ajay-Atul
Song added successfully!

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 3

Forward Playlist:
Kesariya - Arjit [CURRENT]
Heeriye - Jasleen

Zingat - Ajay-Atul

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 4

Reverse Playlist:
Zingat - Ajay-Atul
Heeriye - Jasleen
Kesariya - Arjit

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 5
Now Playing: Kesariya - Arjit

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song

6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 6
Now Playing: Heeriye - Jasleen

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 7
Now Playing: Kesariya - Arjit

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 9
Enter number of repetitions: 2

Repetition 1:
Playing: Kesariya - Arjit
Playing: Heeriye - Jasleen

Playing: Zingat - Ajay-Atul

Repetition 2:
Playing: Kesariya - Arjit
Playing: Heeriye - Jasleen
Playing: Zingat - Ajay-Atul
Repeat completed.

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 2
Enter title to delete: Heeriye
Song deleted successfully!

===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 3

Forward Playlist:
Kesariya - Arjit [CURRENT]
Zingat - Ajay-Atul

```
===== MUSIC PLAYLIST MANAGER =====
1. Add Song
2. Delete Song
3. Display Playlist
4. Display Playlist in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter your choice: 11
Exiting Music Playlist Manager.
PS C:\Users\Admin\OneDrive\Desktop\DSAcce2>
*/
```